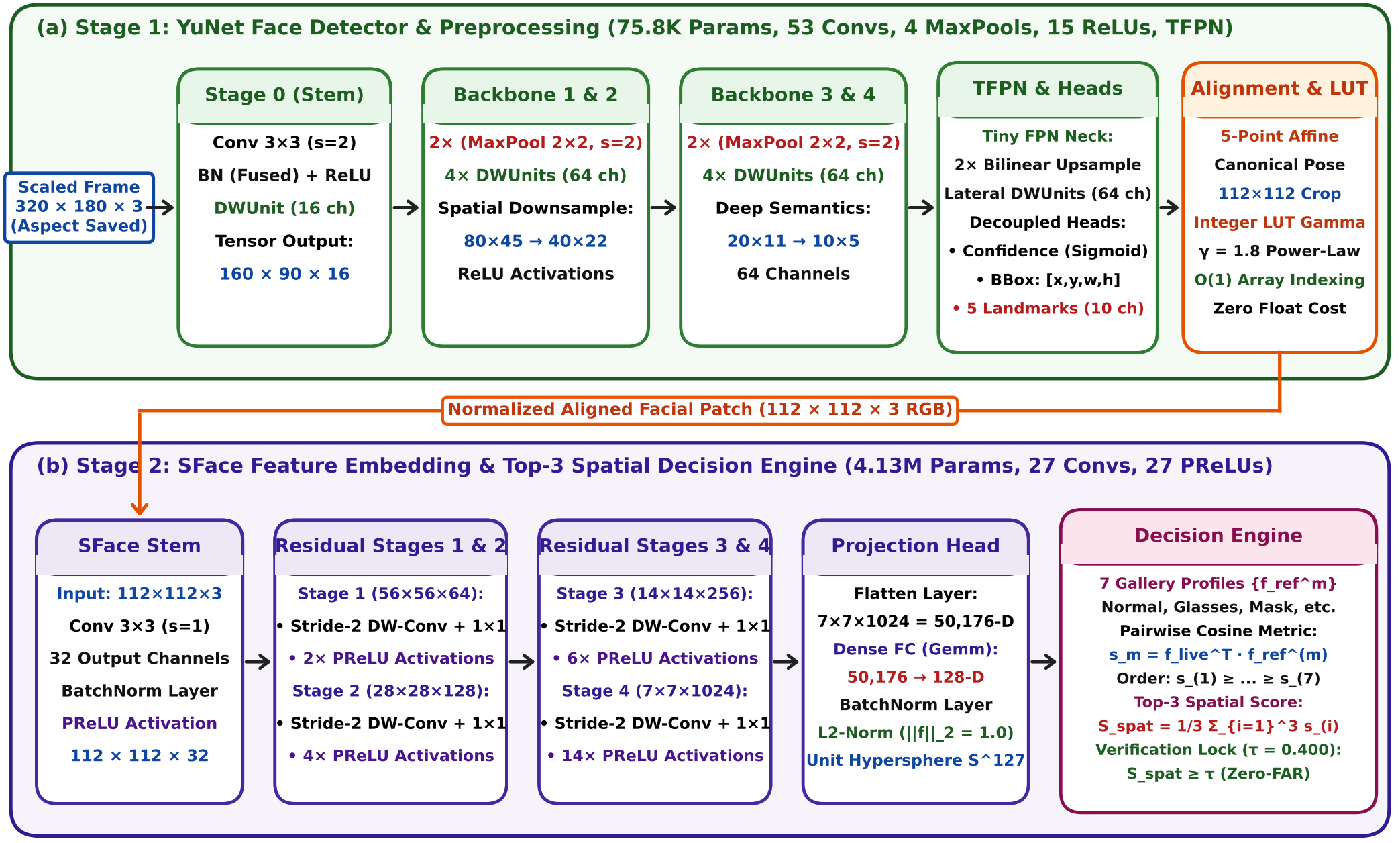
(a) Stage 1: YuNet Face Detector & Preprocessing (75.8K Params, 53 Convs, 4 MaxPools, 15 ReLUs, TFPN)
Scaled Frame
320 × 180 × 3
(Aspect Saved)
Stage 0 (Stem)
Conv 3×3 (s=2)
BN (Fused) + ReLU
DWUnit (16 ch)
Tensor Output:
160 × 90 × 16
Backbone 1 & 2
2× (MaxPool 2×2, s=2)
4× DWUnits (64 ch)
Spatial Downsample:
80×45 → 40×22
ReLU Activations
Backbone 3 & 4
2× (MaxPool 2×2, s=2)
4× DWUnits (64 ch)
Deep Semantics:
20×11 → 10×5
64 Channels
TFPN & Heads
Tiny FPN Neck:
2× Bilinear Upsample
Lateral DWUnits (64 ch)
Decoupled Heads:
• Confidence (Sigmoid)
• BBox: [x,y,w,h]
• 5 Landmarks (10 ch)
Alignment & LUT
5-Point Affine
Canonical Pose
112×112 Crop
Integer LUT Gamma
γ = 1.8 Power-Law
O(1) Array Indexing
Zero Float Cost
Normalized Aligned Facial Patch (112 × 112 × 3 RGB)
(b) Stage 2: SFace Feature Embedding & Top-3 Spatial Decision Engine (4.13M Params, 27 Convs, 27 PReLUs)
SFace Stem
Input: 112×112×3
Conv 3×3 (s=1)
32 Output Channels
BatchNorm Layer
PReLU Activation
112 × 112 × 32
Residual Stages 1 & 2
Stage 1 (56×56×64):
• Stride-2 DW-Conv + 1×1
• 2× PReLU Activations
Stage 2 (28×28×128):
• Stride-2 DW-Conv + 1×1
• 4× PReLU Activations
Residual Stages 3 & 4
Stage 3 (14×14×256):
• Stride-2 DW-Conv + 1×1
• 6× PReLU Activations
Stage 4 (7×7×1024):
• Stride-2 DW-Conv + 1×1
• 14× PReLU Activations
Projection Head
Flatten Layer:
7×7×1024 = 50,176-D
Dense FC (Gemm):
50,176 → 128-D
BatchNorm Layer
L2-Norm (||f||_2 = 1.0)
Unit Hypersphere S^127
Decision Engine
7 Gallery Profiles {f_ref^m}
Normal, Glasses, Mask, etc.
Pairwise Cosine Metric:
s_m = f_live^T · f_ref^(m)
Order: s_(1) ≥ ... ≥ s_(7)
Top-3 Spatial Score:
S_spat = 1/3 Σ_{i=1}^3 s_(i)
Verification Lock (τ = 0.400):
S_spat ≥ τ (Zero-FAR)